

the German aircraft tended to stay on their own side of the trenches and concentrate their resources in ever larger units capable of winning local air superiority on crucial sectors of the front.

Rapid expansion

A battle for production intensified in step with the struggle at the front. The growth in volume of engine and airframe output was spectacular. Very early in the war, military contracts allowed small manufacturers to grow into major industrial concerns. The French company Nieuport was not untypical in seeing its turnover grow from 285,000 francs in 1914 to 26.4 million francs in 1916. New players entered the aero-engine and aircraft industries, notably automobile manufacturers such as Renault and Fiat.

Expansion was most rapid in Britain, which had entered the war with a weak aircraft industry almost entirely dependent on imported French engines. By the end of the war, Britain had the largest aircraft

industry in the world, employing an estimated 270,000 workers.

Plagued by shortages of skilled labor and of vital materials, Germany critically lost out in the battle for volume production. In 1917, the Germans undershot a production target of 1,000 aircraft a month – at a time when the British and French between them were manufacturing about 30,000 aircraft a year.

Quantity was not, of course, the same as quality. Delays and bureaucratic incompetence sometimes led to aircraft being manufactured that were obsolescent before they were ever flown – the notorious R.E.8s and Renault ARs delivered to the front in 1917 were cases in point. The twin goals of maximizing output and improving aircraft performance often

proved contradictory. Aircraft that had been good in their day were kept too long in production so that the demand for numbers could be met.

“Aviation has assumed
a capital importance...
It is necessary to be
master of the air.”

GENERAL HENRI PÉTAIN, MAY 1917

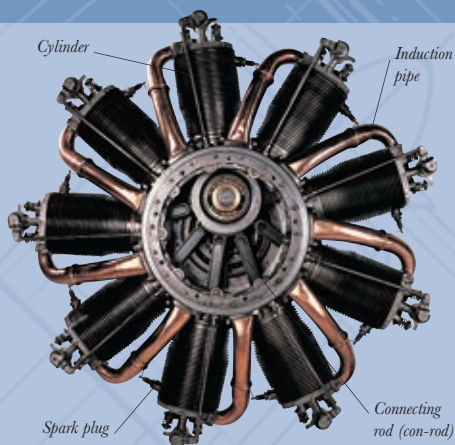
Necessity breeds invention

But there was also a built-in conservatism through the need to exploit existing resources and tried-and-tested techniques. Throughout the war the vast majority of Allied aircraft remained strut-and-wire biplanes, with a fabric skin stretched over a wooden frame. They achieved improved performance largely through the use of more powerful and reliable engines. The Germans were more innovative in their use of materials, partly because of shortages of good quality wood and of skilled workers required to build wooden-frame aircraft. For their fabric-skinned machines, the Germans mostly adopted welded steel-tube

POWERING UP

THE POWER OF ENGINES USED IN combat aircraft grew from around 80hp at the start of the war to a maximum of 400hp by 1918. The two main families of World War I power plant were rotary and in-line water-cooled engines. Rotary engines were lighter and more compact, but ran into problems when required to deliver over 150hp. With their cylinders whirling around a fixed crankshaft they created a powerful gyroscopic effect that made an aircraft tricky to fly, but they worked well on maneuverable dogfighters such as the Sopwith Camel and Fokker Triplane. In-line engines powered stronger, faster aircraft.

The Allies won the battle for engine development because they had a wider range of suppliers, mostly French. The Americans produced the most powerful engine of the war – the 12-cylinder, 400hp Liberty engine – in 1918.



LE RHÔNE 9B ROTARY ENGINE

Many of the rotary engines used by the Allies were produced by the Gnome and Le Rhône companies. Despite some disturbing characteristics, including spraying the pilot with castor oil, they were light and powerful. The best tribute to their quality is that the Germans often installed captured Allied rotary engines in place of their own.

“HISSO” IN-LINE ENGINE

The in-line Hispano-Suiza V8 is often regarded as the outstanding engine of World War I. It was powerful, compact, durable, and light for its size. The engine was used in aircraft like the SPAD XIII and the S.E.5.

